

Modular and predictable assembly of porous organic molecular crystals

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Nanoporous molecular frameworks^{1–7} are important in applications such as separation, storage and catalysis. Empirical rules exist for their assembly but it is still challenging to place and segregate functionality in three-dimensional porous solids in a predictable way. Indeed, recent studies of mixed crystalline frameworks suggest a preference for the statistical distribution of functionalities throughout the pores⁷ rather than, for example, the functional group localization found in the reactive sites of enzymes⁸. This is a potential limitation for ‘one-pot’ chemical syntheses of porous frameworks from simple starting materials. An alternative strategy is to prepare porous solids from synthetically preorganized molecular pores^{9–15}. In principle, functional organic pore modules could be covalently prefabricated and then assembled to produce materials with specific properties. However, this vision of mix-and-match assembly is far from being realized, not least because of the challenge in reliably predicting three-dimensional structures for molecular crystals, which lack the strong directional bonding found in networks. Here we show that highly porous crystalline solids can be produced by mixing different organic cage modules that self-assemble by means of chiral recognition. The structures of the resulting materials can be predicted computationally^{16,17}, allowing *in silico* materials design strategies¹⁸. The constituent pore modules are synthesized in high yields on gram scales in a one-step reaction. Assembly of the porous co-crystals is as simple as combining the modules in solution and removing the solvent. In some cases, the chiral recognition between modules can be exploited to produce porous organic nanoparticles. We show that the method is valid for four different cage modules and can in principle be generalized in a computationally predictable manner based on a lock-and-key assembly between modules.

A basic tool in the synthesis of functional extended solids is the ability to combine different chemical entities in a controlled and modular fashion. This has been demonstrated for structurally related, or ‘isorecticular’, porous metal–organic frameworks (MOFs)³. MOFs can be prepared with more than one chemical function, either by direct reaction of mixed precursors⁷ or by post-synthetic modification¹⁹. Although both MOFs and zeolites can comprise fused, compartmentalized cages, it is still generally challenging to segregate structural units in a programmed and predictable way.

Most nanoporous networks are synthesized in ‘one-pot’ chemical reactions where all of the precursors are mixed together simultaneously^{1–7}. The three-dimensional network structure arises from self-assembly of the components. By contrast, natural products are synthesized in stepwise reaction sequences where isolable molecular intermediates are elaborated and combined to create more complex structures. An analogous, supramolecular strategy²⁰ for porous organic solids would be to preorganize larger chemical subunits, or pore modules, before assembling the extended crystal. This approach requires building blocks, or tectons^{21,22}, that are self-assembling,

prefabricated molecular analogues of the secondary building units in networks such as MOFs^{1–5,7}. To be broadly useful, the pore modules should pack together in predictable ways. Individual modules could then be designed to incorporate desirable chemical functionalities, either by chemical derivatization¹⁵ or by physical encapsulation within the molecular pores²³. Mixing different functional modules might produce porous solids with unusual properties, perhaps, for example, by combining both acid- and base-containing cage modules within the same porous solid along with vacant, flow-through pores. In practice, however, many components of this strategy are currently missing. Although a large number of porous molecular solids are known^{9–15}, as highlighted in a recent review²⁴, the rules that underpin their three-dimensional, non-covalent assembly are poorly understood. In this respect, the notion of ‘supramolecular synthesis’²⁰ is still unfulfilled. Levels of porosity in such molecular organic solids are also modest: until recently^{12–15}, Brunauer–Emmett–Teller specific surface areas, S_{ABET} , of less than $400 \text{ m}^2 \text{ g}^{-1}$ were typical²⁴. Moreover, porous molecular solids could not be described as modular because almost all examples are single-component crystals.

In this study, we report the production of porous organic molecular co-crystals, thus demonstrating a new modular assembly concept. We also describe computational methods to predict these crystal structures *ab initio*, greatly enhancing the long-term prospects for rational materials design¹⁸. The materials were fabricated from combinations of the four pore modules shown in Fig. 1a. The first porous co-crystal was constructed from two organic cages that we described previously¹²: cage 1 and cage 3-*R*. Porosity is covalently prefabricated in the individual tetrahedral cage molecules such that each module has four triangular pore windows with diameters of around 6 Å (Fig. 1a; see also scheme 1 in Supplementary Information). Each cage is just over 1 nm in size. Both cage modules have helical chirality: 1 comprises, in crystalline form, an equimolar mixture of the helical enantiomers 1-*S* and 1-*R*^{12,14}, whereas 3-*R* is homochiral. Both cages are soluble in common solvents and can be simply mixed together in solution. Slow evaporation of an equimolar solution of 1 and 3-*R* did not lead to separate crystals of the individual modules, but rather to a new single-phase crystalline material. Remarkably, the material is a quasisracemic co-crystal²⁵, (1-*S*, 3-*R*). That is, it consists exclusively of the *S* helical enantiomer of 1 crystallized with 3-*R* (Fig. 1b). The apparent loss of the 1-*R* enantiomer, despite 100% sample mass recovery from crystallization, is explained by variable-temperature ¹H NMR measurements. This shows that the helical configurations of 1 interconvert rapidly in solution¹⁴ (Supplementary Fig. 2). The chirality of 1 is therefore dynamically resolved on crystallization with the homochiral cage, 3-*R* (Fig. 2a). Cage 1 is an ‘amphichiral’ module: it can also pair with 3-*S* to form the opposite quasisracemic co-crystal, (1-*R*, 3-*S*). As discussed below, however, this assembly strategy is not limited to dynamically chiral molecules.

The crystal packing for (1-*S*, 3-*R*) is also shown in Fig. 2: the 1-*S* and 3-*R* modules alternate in the crystal lattice in a face-centred cubic

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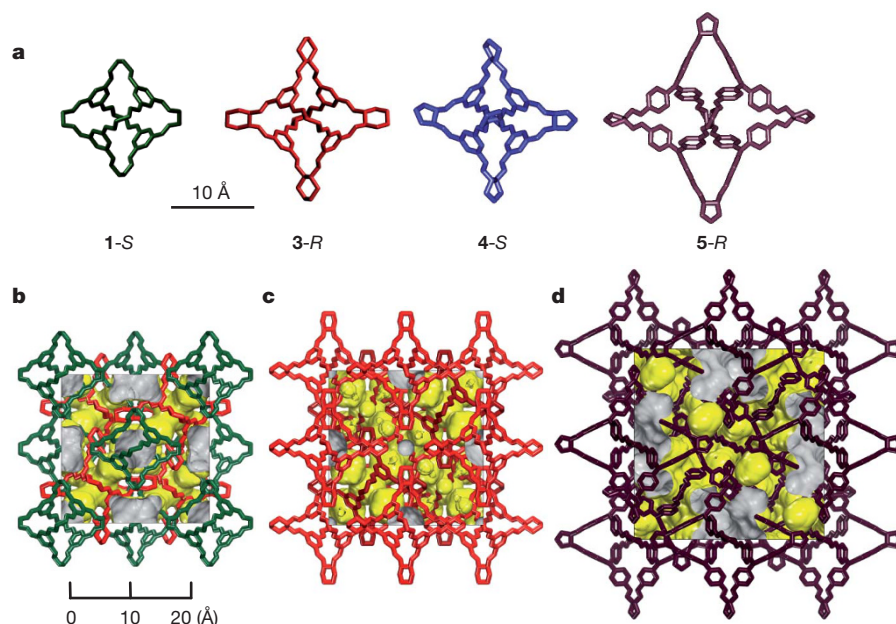


Figure 1 | Modular assembly of porous organic cages. **a**, Structures of four organic cage modules (hydrogen atoms omitted for clarity). Cage 1 is shown as the *S* enantiomer but this module is amphichiral and can interconvert between the *R* and *S*

forms. **b–d**, Crystal structures of porous organic solids formed from these modules, with the Connolly surface shown in yellow (probe radius, 1.82 Å): quasiracemic co-crystal (1-*S*, 3-*R*) (**b**), racemic crystal (3-*S*, 3-*R*) (**c**) and chiral crystal 5-*R* (**d**).

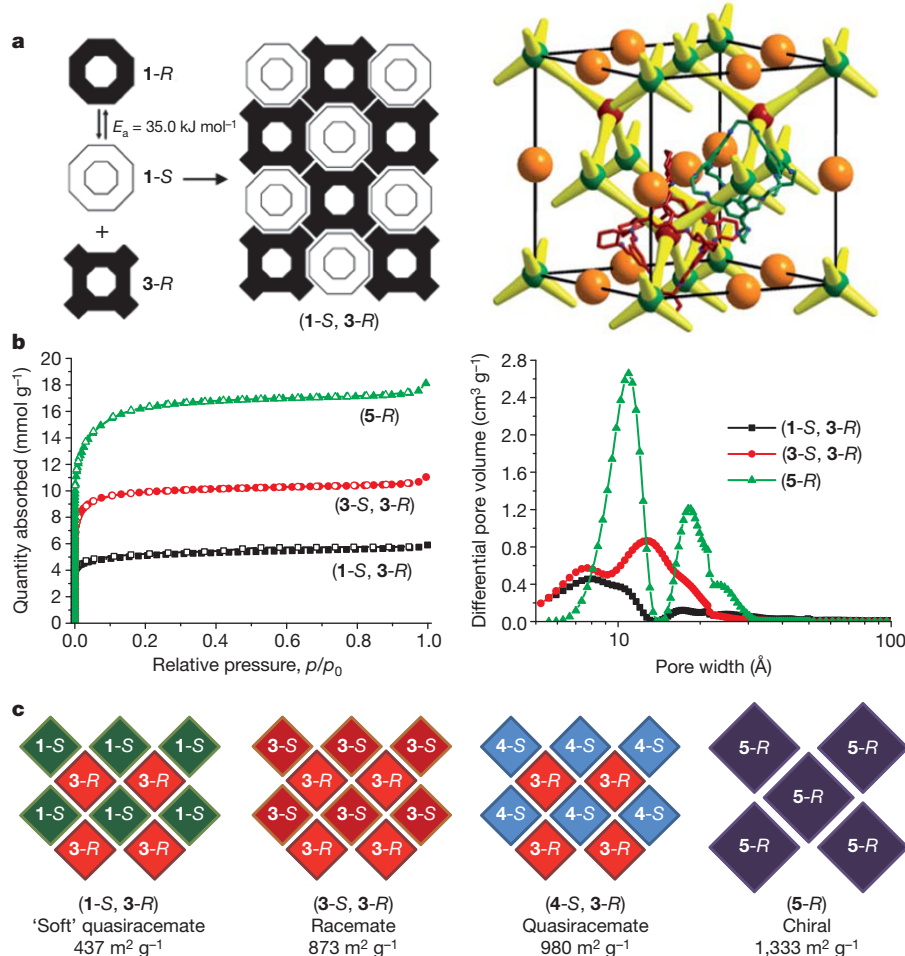


Figure 2 | Window-to-window assembly results in porosity. **a**, Helical chirality in 1 is dynamically resolved by heterochiral co-crystallization with 3-*R*. The schematic packing diagram for (1-*S*, 3-*R*) shows the centres of modules 1-*S* and 3-*R* as green and red spheres, respectively; orange spheres represent interstitial voids that are not connected to the diamondoid pore network, which is illustrated in yellow. E_a is the activation energy for conversion

between 1-*S* and 1-*R*, as measured by variable-temperature NMR. **b**, Nitrogen gas sorption analysis for crystals and co-crystals shows that pore volume and pore size can be varied systematically, as in isorecticular networks. Filled and open symbols represent sorption and desorption isotherms, respectively. p_0 , atmospheric pressure. **c**, Scheme showing packing for various crystals and co-crystals.

arrangement, analogous to the ZnS ‘zinc blende’ structure. Each cage forms window-to-window interactions with four partner cages of the other type. The result is an interconnected diamondoid pore network. No polymorphs of pure **1** have been found that pack in a window-to-window fashion^{12,14}. Therefore, this packing mode is directed by the presence of the chiral co-module, **3-R**. The window-to-window packing arrangement creates permanent micropore channels in the co-crystal, which has a type-I nitrogen sorption isotherm at 77 K (Fig. 2) and a specific surface area of $S_{\text{BET}} = 437 \text{ m}^2 \text{ g}^{-1}$. Like the other materials described here, the co-crystal is stable to desolvation and has good thermal stability, showing little weight loss until the onset of decomposition at 350 °C (Supplementary Fig. 7).

The heterochiral pairing (**1-S**, **3-R**) can be considered a directional tecton²², comparable to reversible supramolecular interactions such as hydrogen bonding²¹ and the ‘sextuple aryl embrace’²⁶ that involves interlocking aryl rings. Density functional theory (DFT) calculations for isolated cage pairs indicate that the heterochiral window-to-window interaction is 18 kJ mol^{-1} more stable than the equivalent homochiral interaction and much more stable than other hypothetical window-to-arene or arene-to-arene pairs that would lead to disconnected pores (Supplementary Fig. 8). Lattice energy calculations confirm that this heterochiral pairing preference carries over to the solid

state and, more significantly, that the observed co-crystal structure can be predicted *ab initio* from the molecular formulae of the modules. Calculations using Monte Carlo simulated annealing to generate hypothetical (**1-S**, **3-R**) crystal structures, followed by energy minimization using anisotropic atom–atom potentials^{27,28}, showed the observed packing mode for (**1-S**, **3-R**) to be the global lattice energy minimum (Fig. 3), with good agreement between the *ab initio* predicted structure and the experimental single-crystal X-ray structure (Fig. 3b). The most stable hypothetical homochiral (**1-R**, **3-R**) structure, which lacks window-to-window packing, was predicted to be 18.8 kJ mol^{-1} less stable than the observed quasiracemate, (**1-S**, **3-R**). These calculations therefore rationalize the preference for **1** to adopt the **1-S** configuration in the co-crystal and to pack in a window-to-window fashion: that is, both the preferred chirality and the resultant porosity in the solid can be predicted *ab initio*. To verify the atom–atom-potential lattice energy calculations, we performed solid-state DFT calculations on the observed quasiracemate and low-energy predicted homochiral structures: these calculations confirmed the preference for heterochiral packing.

This behaviour is not limited to the pairing of **1-S** and **3-R**. The enantiomers **3-S** and **3-R** also strongly prefer heterochiral window-to-window pairs and assemble in that fashion in a (**3-S**, **3-R**) racemic

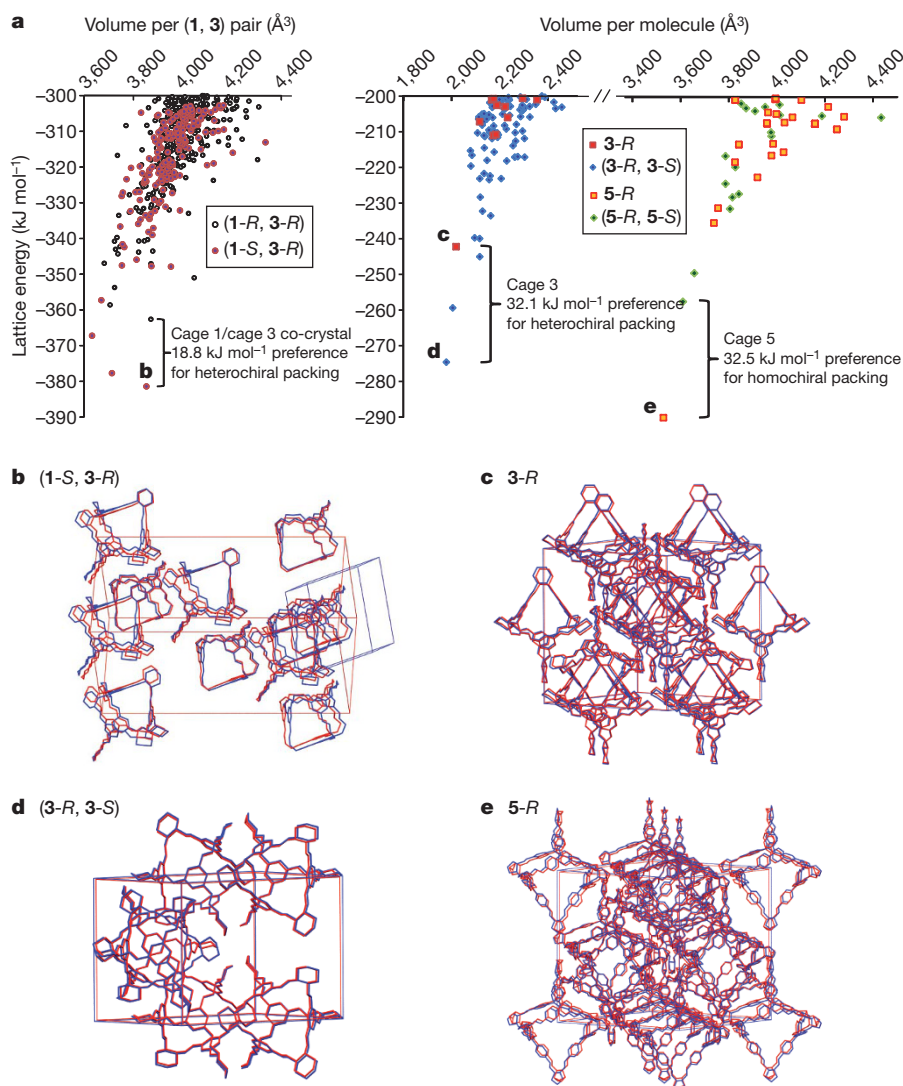


Figure 3 | Three-dimensional cage assembly can be predicted computationally. **a**, Lattice energy rankings rationalize the heterochiral packing preference for the (**1**, **3**) co-crystal (structure **b** is favoured over all hypothetical homochiral predicted structures), the racemic packing preference for cage **3** (structure **d** is favoured over **c**) and the chiral preference for

5 (structure **e** is favoured over all hypothetical racemates). **b–e**, Packing diagrams show the excellent fit between the calculated global-minimum structures (blue) and the experimentally determined structures (red). The predicted (**1-S**, **3-R**) structure in **b** is slightly less symmetrical than the observed **R3** space-group symmetry. The **P1** unit cell is shown.

crystal (Fig. 1c) to give a porous solid with $SA_{\text{BET}} = 873 \text{ m}^2 \text{ g}^{-1}$. In this case, the chirality in both modules is fixed rather than dynamic. As before, DFT simulations suggest a significant energy gain (19 kJ mol^{-1}) in the formation of heterochiral dimers. Again, the crystal structure can be predicted *ab initio*. The experimentally observed racemic packing is the global energy minimum in the set of predicted crystal structures, and there is close agreement between the predicted and observed structures (Fig. 3d). These calculations also suggest a global preference for heterochiral packing modes rather than homochiral. A large energetic gain, of 32 kJ mol^{-1} , is calculated for the (3-S, 3-R) racemic crystal over the most stable predicted homochiral structure for 3-R. The global-minimum homochiral prediction also closely reproduces the observed structure for 3-R¹² (Fig. 3c), which, unlike **1**, can be obtained from enantiopure solutions because 3-R does not interconvert with its enantiomer. As for the (1-S, 3-R) co-crystal, the atom–atom lattice energy calculations were verified using periodic DFT calculations, which resulted in similar calculated energy differences (Supplementary Table 1 and Supplementary Fig. 13). An analogous set of experimental observations and crystal structure predictions was obtained for a new cage module, 4-S, which has cyclopentane vertices rather than cyclohexane (Fig. 1a). This module forms a quasiracemic co-crystal, (4-S, 3-R), with $SA_{\text{BET}} = 980 \text{ m}^2 \text{ g}^{-1}$. In this case, the predicted global-energy-minimum crystal structure is an ordered version of the most probable site-disordered space-group-*F4*/32 structure, according to powder X-ray data (Supplementary Figs 12 and 15). By itself, 4-S does not pack in a window-to-window fashion (Supplementary Fig. 18). Hence, like (1-S, 3-R), this packing mode is directed by the partner module, 3-R.

Not all systems favour heterochiral assembly and this, too, is predictable from the calculated crystal energy landscape. A new module, 5-R (Fig. 1a), was synthesized by the [4 + 6] cycloaddition reaction between tri(4-formylphenyl)amine and the chiral diamine (*R,R*)-1,2-cyclopentanedi-amine. Cage 5-R is substantially larger than modules **1**, **3** and **4**. For example, the tetrahedron inscribed by the centres of the triangular faces of 5-R has a volume that is 3.8 times larger than the comparable tetrahedron for cage **1** (Supplementary Fig. 24). In this case, lattice energy calculations suggest homochiral window-to-window packing as the clear global energy minimum, and this predicted structure is observed experimentally for 5-R (Fig. 3e); again, DFT calculations agree broadly with the energy differences obtained by atom–atom-potential lattice energy calculations. To our knowledge, 5-R ($1,702 \text{ g mol}^{-1}$) is the largest organic molecule to be successfully tackled by crystal structure prediction¹⁶. Numerous experiments involving crystallization from mixtures of the modules 5-R and 5-S all led exclusively to homochiral crystals, in agreement with the predicted lattice energy preference over all hypothetical racemic structures. The crystalline solid 5-R has larger pores (compare Fig. 1b, c with Fig. 1d) and a greater pore volume ($0.63 \text{ cm}^3 \text{ g}^{-1}$) than any of the materials produced from the smaller cages **1**, **3** and **4**. The surface area of 5-R ($SA_{\text{BET}} = 1333 \text{ m}^2 \text{ g}^{-1}$) exceeds all but one¹⁵ of the porous molecular (non-network) crystals reported so far^{9–14,24} and is comparable with the first generation of covalent organic frameworks⁶. This larger cage shows that it is possible to prepare molecular organic crystals with bespoke pore sizes, analogous to the well-known series of isorecticular MOFs³ where pore size is defined by organic strut length. A future challenge will be to generalize this non-covalent assembly methodology. Non-identical molecules do not, as a general rule, co-crystallize, and it may be necessary to incorporate specific complementary functionality to induce co-crystallization of dissimilar modules.

We have shown that porous cages can assemble in a modular fashion and, moreover, that the mode of assembly can be predicted accurately using lattice energy calculations. These particular structures are amenable to computation because the directional interlocking of neighbouring cages leads to large energy differences between hypothetical structures. By contrast, most other organic molecules give rise to many

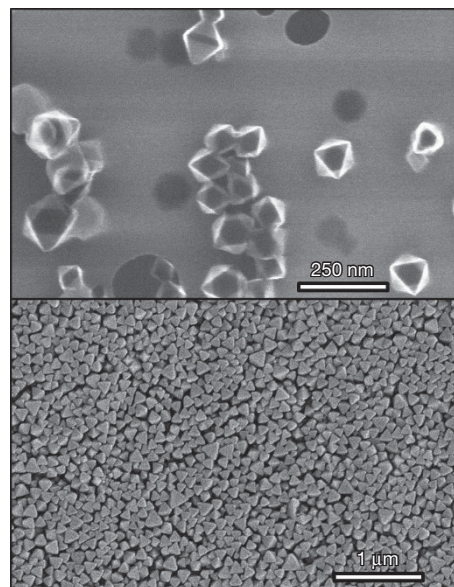


Figure 4 | Module assembly in solution can be used to produce porous nanoparticles. Mixing solutions of 3-S and 3-R leads to rapid precipitation of racemic octahedral nanocrystals of (3-S, 3-R) with an average diameter of 130 nm ($SA_{\text{BET}} = 873 \text{ m}^2 \text{ g}^{-1}$). The micrographs show the same sample imaged at two different magnifications.

distinct possible crystal structures that differ in energy by only a few kilojoules per mole^{16,17}. Larger, conformationally flexible cage modules would be more challenging for these prediction methods, but significant recent advances have been made in dealing with molecular flexibility^{29,30}. Thus, the work presented here opens the way for *in silico* prediction of structure and properties for new candidate porous materials based solely on two-dimensional chemical sketches, thus allowing ‘design by computational selection’.

The solution processability of the cage modules also means that the assembly approach can be extended to achieve structural control beyond the molecular length scale. For example, the (3-S, 3-R) racemate is at least ten times less soluble than the homochiral modules, 3-S and 3-R, and this leads to spontaneous precipitation on mixing of solutions of the two enantiomers (Supplementary Movie 1 and Supplementary Fig. 27). Well-defined, porous (3-S, 3-R) nanocrystals are formed (Fig. 4), thereby translating intermolecular heterochiral tecton interactions into nanoscale morphology control. Porous nanocrystals might make particular applications of these solids possible in future, for example in chiral catalysis or separations.

METHODS SUMMARY

Synthesis of compounds. Cage **1**, cage 4-R and cage 3-R were synthesized in a [4 + 6] cycloaddition reaction involving triformylbenzene and the diamines ethylenediamine, (1S,2S)-cyclopentanedi-amine and (1R,2R)-cyclohexanedi-amine, respectively, using an improved synthetic procedure which produces higher yields than that reported previously¹² (Supplementary Information). Cage 5-R was synthesized by the [4 + 6] cycloaddition reaction between tri(4-formylphenyl)amine and (*R,R*)-1,2-cyclopentanedi-amine. Co-crystals were grown from equimolar solutions of the partner cage modules. Details of the crystallographic analysis, crystal data and gas sorption analysis are described in Supplementary Information.

Crystal structure prediction. Crystal structures were generated in the most commonly observed space groups using a Monte Carlo simulated annealing search method. The lowest-energy structures from the Monte Carlo search were then lattice-energy-minimized using anisotropic atom–atom potentials within the crystal structure modelling software DMACRYS²⁸. Molecular geometries, generated by DFT single-molecule optimization, were treated as rigid throughout the predictions. Further details are given in Supplementary Information.

DFT calculations. Cage pairs and crystal structures were fully optimized in the mixed Gaussian and plane-wave code CP2K³¹, using the TZVP-MOLOPT basis set in combination with Geodecker–Teter–Hutter pseudopotentials and a plane-wave

cut-off of 400 Ry. Molecular and solid-state calculations used the BLYP and PBE functionals, respectively, both with Grimme's D3 dispersion correction³².

Received 22 October 2010; accepted 18 April 2011.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

Acknowledgements We thank EPSRC (EP/H000925/1) and the Dutch Polymer Institute for funding. The UK national high-performance computing service HECToR was used for this work through the HPC Materials Chemistry Consortium (EPSRC grant EP/F067496. A.I.C. and G.M.D. are Royal Society Wolfson Merit Award holder and A.T. and G.M.D. are Royal Society University Research Fellows.

Author Contributions J.T.A.J. prepared cage modules and co-crystals thereof, contributed to powder X-ray diffraction analysis, carried out high-throughput co-crystallization screening experiments and generally coordinated the experimental work; T.H. prepared cage modules and co-crystals and the nanocrystals of (**3-S**, **3-R**), and carried out microscopy; J.T.A.J. and T.H. did the gas sorption analysis; X.W. synthesized module **5**; D.J.A. contributed to the cage synthesis; J.B., M.S., S.Y.C. and A.S. performed the crystallography; K.E.J., A.T., F.C. and B.S. performed molecular and periodic simulations, in particular the DFT studies; G.M.D. led the crystal structure prediction; and A.I.C. designed the project and wrote the paper with contributions from all co-authors.

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